

January 2024, CSE 106
Online on Dynamic Programming (A1/A2)
Time: 40 Minutes

Suppose your friend is an amateur robber planning to rob houses along a street. Each house has a certain amount of money stashed. If your friend robs a house he will have to take all the money stored in the house. There are two constraints stopping your friend from robbing each of the houses. First, your friend can only carry upto a maximum weight. The second one is that adjacent houses have security systems connected, and **it will automatically contact the police if two neighboring houses are broken into on the same night.**

You are given two integer array `nums` and `weights` representing the value and weight of the money in each house, respectively, and a maximum weight your friend can carry. Design and implement an efficient solution with dynamic programming to determine *the maximum amount your friend can rob **without alerting the police.***

Input / Output:

The first two input integers indicate the number of houses n and the maximum weight w that your friend can carry. Then there will be n integers indicating the value stored in each house. Finally, there will be n integers that indicate the weights of the money stored in each house.

Print *the maximum amount your friend can rob **without alerting the police.***

Sample Input	Sample Output
4 50 60 100 120 100 10 20 30 20	200
5 50 60 120 80 100 70 10 20 30 20 10	220

Submission Guidelines: Save your `.cpp` file as `2205<3-digit-Student-ID>.cpp` and upload it on Moodle in the corresponding submission link.